



Hands-on Lab: Leverage AI Tools to Develop a Product

Estimated duration: 30 minutes

Introduction

A technology start-up company employs you in a large city. This city does not have a reliable traffic management system. The current reporting is manual, dependent upon cameras, and slow to update. Your firm believes it can develop an AI-driven system that will monitor and improve traffic flow in the city and provide a return on investment (ROI). A Product Concept Document was developed.

Objectives

- Review the Product Concept Document
- Define how AI can support the product concept
- List how this product could be potentially commercialized
- Develop a product vision

Prerequisites:

Ensure you have viewed the following:

- Video: Stages of AI Product Development
- Video: AI Integration in Product Management Lifecycle
- Reading: AI Support for Product Management Success

You must have access to Microsoft Word, either through the desktop or online version. If you want to sign up for Microsoft Office 365 and need help, view the [Sign up for Microsoft 365 lab](#).

Task 1: Review the Product Concept Document

Develop an AI-Enhanced Traffic Management System	
Entry	Considerations
Market Problem or Opportunity	• Traffic continues to present challenges for area commuters. • Commuters need a tool that allows them to reduce travel time, determine the best mode of transportation, and receive real-time traffic update notifications. • Product must be easy to access, mobile, and cost-effective.
Market Segments	• Product will support commuters, organizations, and the local municipal government agencies requiring traffic management data. • Product can also be used by the local Department of Transportation (DOT) to identify traffic-related problems and traffic management opportunities and prioritize transportation initiative improvements.
Key Financials	• Product development estimate: \$3.5M • Return on investment: 75%
Market Window	• System is required today. • Goal: Design and develop over a 6-month period.
Competitive Landscape	• Project Green Light is already operational in 12 cities worldwide. The project aims to improve traffic flow, save fuel, and reduce emissions. • Manual traffic reporting systems are available on the web.
Main Features and Functionality	• Analyze traffic conditions on a real-time basis and provide route planning guidance • Load trip history and enable retrieval in seconds • Send immediate notifications when there are sudden changes such as mudslides, accidents, construction, etc. • Recommend best method of transportation to reach a destination to include auto, light rail, and

	- bus - Track accidents and recurring issues. Perform root cause analysis and recommend solutions - Perform internal system updates based on new data without human intervention or programming - Enable users to grade their experience and provide feedback to improve system performance - Be capable of interfacing with current DOT infrastructures - Be compatible with current state and federal warning systems - Guard user privacy - Conform to all regulations
Key Differentiators	- App driven. Easy download and compatible with all Apple and Android systems - Optimal personalized route planning - Real-time notification and alert systems - Eco-friendly options - Leverages Intelligent Traffic Systems (ITS) - Dynamic parking solutions - Exceeds "Project Green Light" capabilities - Integrates and enhances existing systems
Go-to-Market Logistics	- Develop initial application (6 months) - Launch marketing campaign - Reach 200,000 Users within 6 months - Integrate real-time data-driven updates daily
Business Success Measurements	- Design and development complete in 6 months 200,000 users within 6 months - Stay within budget of \$3.5M - ROI of 75% in year one - Customer satisfaction rating of 4.7

Task 2: Download the Product Concept Evaluation Worksheet

Keeping the CTRL key pressed, click to download the [Product Concept Evaluation Worksheet](#) that provides tables to complete this assignment.

Note: Use the COMMAND key instead of CTRL on Mac systems.

Task 3: List at least three ways AI can be used to achieve product objectives

- Define product objectives where AI can improve viability of features and functions.

Task 4: List at least three ways this product could be effectively commercialized

- Commercialization is defined as "Manage or exploit in a way designed to make a profit."
- How can this firm make a profit from launching this product?

Task 5: Develop a product vision

- Develop a product vision to support this product.

Task 6: Review the solution

Keeping the CTRL key pressed, click to download the [Product Concept Potential Solution](#).

Note: Use the COMMAND key instead of CTRL on Mac systems.

- Compare your analysis to a recommended solution.
- Update your analysis as required.

Note: Your solution does not need to match this solution. You may identify items not included in the recommended solution. That is fine.